

RUNG 1 · BASELINE

Historical Log Replay

Offline Dataset Metrology

Gate 0: Affordance Error < 2 cm

Execution Substrate:

- Static logged sensor streams
- Offline GPU training server
- Open-loop prediction loss

Verification Focus:

- Model affordance accuracy
- Latent token prediction error
- Offline inference throughput

Blind Spot / Limit:

- Zero closed-loop causality
- Blind to physical momentum
- Misses P_{99} latency tails

RUNG 2 · CLOSED LOOP

Physics Simulation

Domain Randomization

Gate 1: > 98% Pass (10^4 Seeds)

Execution Substrate:

- MuJoCo / Isaac Sim physics
- Multi-GPU parallel rollouts
- Randomized mass, friction, light

Verification Focus:

- Closed-loop policy stability
- Dynamic collision avoidance
- Generalization over 10^4 seeds

Blind Spot / Limit:

- Sim-to-real physics gap
- Idealized compute/bus timing
- Misses silicon fault modes

RUNG 3 · TARGET SILICON

Hardware-in-the-Loop

Real Silicon + Plant Emulator

Gate 2: 100% Fault Containment

Execution Substrate:

- Production MPU + Real-Time MCU
- 10 kHz FPGA plant emulator
- Real SPI, CAN, and MIPI buses

Verification Focus:

- Seeded cross-layer faults
- MCU safety watchdog timing
- Memory DMA bus contention

Blind Spot / Limit:

- Synthetic visual assets
- Emulated optical lighting
- Bounded human interaction

RUNG 4

Shadow Fleet Mode

Passive Real-World Fleet

Gate 3: 0 Divergence / 500 hrs

Execution Substrate:

- Physical robot in active fleet
- Real-world warehouse ODD
- Inferences run in background

Verification Focus:

- Uncontained divergence audit
- Long-tail corner case capture
- Zero actuator safety authority

Blind Spot / Limit:

- Policy cannot actuate world
- Dependent on fleet coverage
- Requires data privacy filters

THE PROMOTION PRINCIPLE: A high score on Rung 1 or 2 is a necessary prerequisite but never sufficient proof of release. Only Rungs 3 and 4 provide defensible physical evidence for the **Deploy / Condition / Refuse** release verdict.